GRADUATION PROJECT · 2026

CLARA

AI-Powered System to manage legal offices work

✦ Smart · 📄 OCR · ◇ NLP

SUPERVISOR
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THE CHALLENGE

Problem Statement

01

Task Management

Work is scattered across chats and calls with no structured way to assign or track it.

02

Tracking Employee Work

Managers lack visibility into who is working on what, across branches and departments.

03

Storing Hard Copies

Thousands of paper contracts pile up with no organized way to archive or retrieve them.

AI FEATURES

04

Editing Contracts

Amending legal documents by hand takes hours and invites costly human error.

THE CHALLENGE

Problem Statement: Task Management

THE OLD WAY

Work is scattered across WhatsApp chats, calls, and informal messages with no structured way to assign, prioritize, or track it from request to completion.

WhatsApp

Phone calls

Group chats

Verbal notes

WITH CLARA

Every task lives in one place — assigned, prioritized, and tracked automatically from request to completion.

Contract Review — Client A

Assigned to Sara K.

Due in 2 days On track

THE CHALLENGE

Problem Statement: Track Employee Status

THE OLD WAY

12+
BRANCHES

340+
EMPLOYEES

Who's working?

Who's on leave?

Task progress?

Hours logged?

WITH CLARA

One Place, Full Visibility

- Interactive employee status dashboard
- Live task & company status board
- Employee directory with search & filters
- Full profile page per employee

THE CHALLENGE

Problem Statement: Organizing Files

THE OLD WAY

- Hard-copy archives
- Personal laptop copies
- Flash drives
- Outdated versions

WITH CLARA

Centralized File System

- Organized by company
- Organized by document type
- Instant search across documents
- Always the latest version

THE CHALLENGE

Problem Statement: Editing Contracts

MANUAL EDITING		✦ WITH CLARA AI
45 min per contract edit	→	**3 min** per contract edit
6 contracts / 5-hour workday	**15x faster** **16x output**	**100** contracts / 5-hour workday

AI

POWERING THE NEXT GENERATION OF CLARA

THE AI ENGINE

What Powers CLARA

MODEL 01

CAT

Edits Arabic legal contracts with precision and speed.

MODEL 02

CORE

A trained OCR model for information extraction from pdf contractes.

SYSTEM 03

RAG System

Retrieves relevant legal context to ground every response.

CAT

CAT's purpose is to edit Arabic legal contracts — automatically, accurately, and at a fraction of the time it takes by hand.

MODEL FOCUS Arabic Contract Editing

CAT — DATASET

Training Data

465 total data points

375 train | 45 validation | 45 test

● Train ● Validation ● Test

SAMPLE JSON STRUCTURE

```
{
  "action_title": "string",
  "before": [ "list of strings" ],
  "after": [ "list of strings" ],
  "prompt": "string"
}
```

CAT — MODEL SELECTION

Choosing the Base Model

Falcon	LLaMA	Gemma	Qwen
✕ REJECTED	✕ REJECTED	✕ REJECTED	✓ SELECTED

Qwen was chosen for its **superior structural consistency** and ease of fine-tuning compared to the alternatives.

CAT — MODEL SELECTION

Picking a Qwen Version

KEPT

Qwen 2.5

Text-only, stable, well-documented — solid baseline for fine-tuning.

KEPT

Qwen 3

Newer generation, text-only, improved reasoning structure.

REJECTED

~~Qwen 3.5~~

Multi-modal — includes vision layers we don't need. Removing them strips its advantage, making it pointless for our text-only task.

CAT — MODEL SELECTION

Size Variants per Version

Qwen 2.5

0.5B PARAMS	1.5B PARAMS	3B PARAMS	7B PARAMS

Qwen 3

0.6B PARAMS	1.7B PARAMS	4B PARAMS	8B PARAMS

CAT — TRAINING

From Base Models to Candidates

Unsloth
Training framework

→

LoRA Fine-Tuning
Applied to all 8 size variants

→

8 Trained Models
4 sizes × 2 Qwen versions

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From Base Models to Candidates

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8 Trained Models
4 sizes × 2 Qwen versions

Which one is best? We need a **score function** to decide.

The Score Function

METRIC	WHAT IT MEASURES	METHOD
ROUGE-L	Text similarity	**rouge-score** (Google) — LCS-based overlap
Semantic Similarity	Meaning match	**Qwen3-embeddings-0.6B**, cosine similarity
Edit Similarity	Character-level match	**difflib SequenceMatcher** (Ratcliff/Obershelp)
Change Accuracy	Number match	**Regex + digit normalizer** checking key numbers

CAT — SCORE FUNCTION

Weighting the Metrics

FINAL SCORE FORMULATION

$$F = 0.45(CA) + 0.30(SS) + 0.15(\text{ROUGE-L}) + 0.10(ES)$$

CAT — SCORE FUNCTION

Metric Equations

ROUGE-L

$$F = 2PR / (P + R)$$

$P = LCS(pred, gold) / |pred|$ · $R = LCS(pred, gold) / |gold|$

Semantic Similarity

$$sim = (e\_pred \cdot e\_gold) / (\|e\_pred\| \|e\_gold\|)$$

Cosine similarity × 100, using e5-base embeddings

Edit Similarity

$$ratio = 2M / T$$

M = matching characters, T = total characters (difflib)

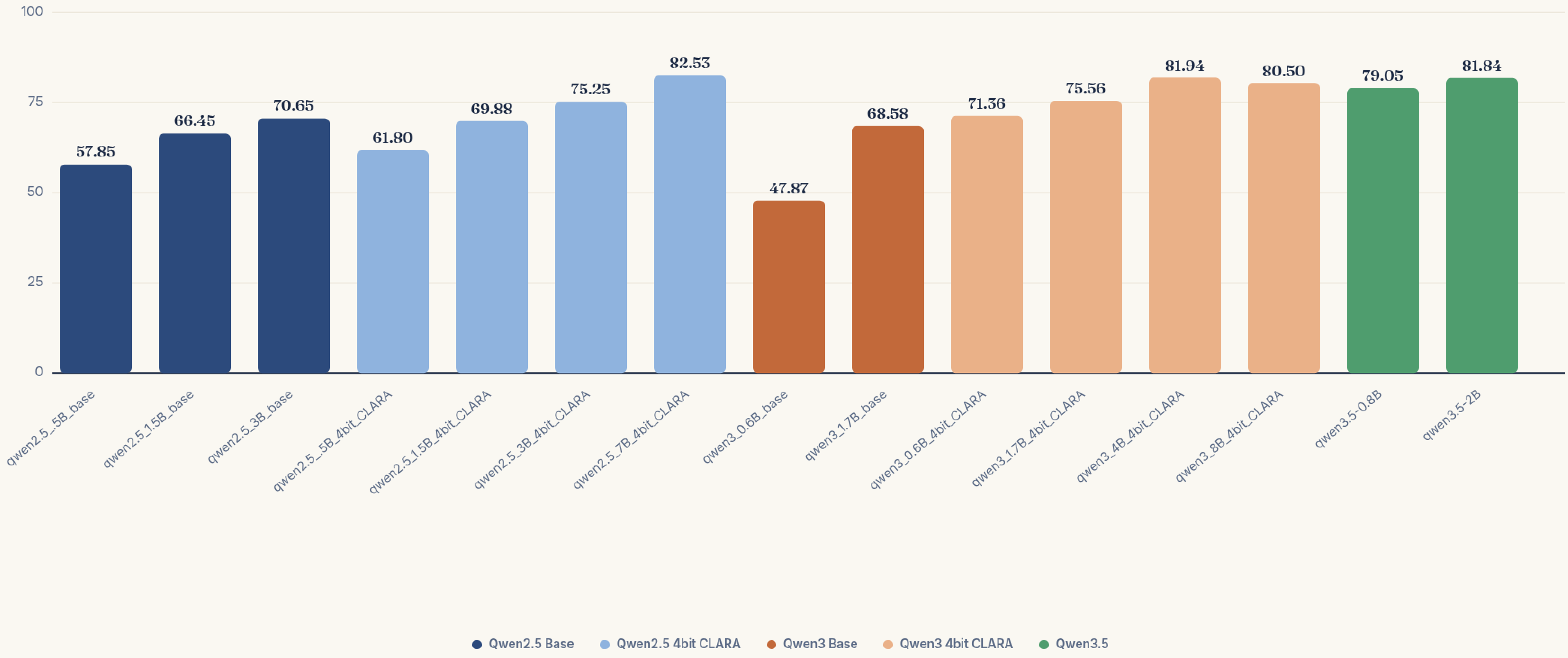
Change Accuracy

$$Recall = |Nums\_pred \cap Nums\_gold| / |Nums\_gold| \times 100$$

Checks whether key numbers were reproduced correctly

CAT — MODEL SELECTION

Final Scores Across 15 Candidates



CORE

> CORE is a trained OCR model for extracting information from PDF contracts — turning scanned and digital documents into structured, usable data.

MODEL FOCUS **OCR & Information Extraction**

CORE — DATASET

Source Contracts

30
PDF contracts — Train

2
PDF contracts — Test

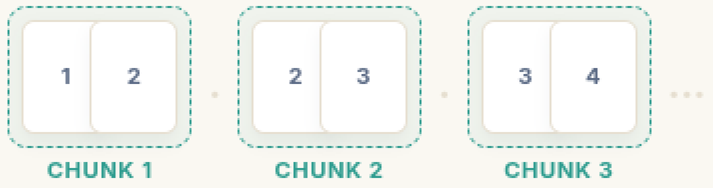
32 total contracts, split **30 / 2** before any preprocessing begins.

CORE — DATASET

Preprocessing Pipeline

1. Each PDF page is converted into a JPG image.

2. Consecutive JPGs are grouped into overlapping 2-page chunks.



3. 440 total chunks, then expanded with augmentation.

CORE — DATASET

Final Dataset

1,640 total sample points (440 chunks → augmented)

SAMPLE JSON STRUCTURE

```
{
 "chunk_i": [ jpg_j, jpg_k ],
 "company_name": "string",
 "company_address": "string",
 "company_type": "string",
 "law_reference": {
  "law_number": "string number",
  "law_year": "string number"
 },
 "company_capital": "string",
 "commercial_register_number": "string",
 "approval_date": "string",
 "real_estate_registry_minutes_number": "string",
 "company_duration": "string",
 "company_activity": "string",
 "articles": [
  { "id": "0", "value": "string" },
  { "id": "1", "value": "string" }
  // ... rest of articles
 ]
}
```

CORE — MODEL SELECTION

Choosing the Base Model

Gemma

✕ REJECTED

Heavy hallucinations,
especially with Arabic text

DeepSeek

✕ REJECTED

Heavy hallucinations,
especially with Arabic text

2-Model System

✕ REJECTED

Running cost too high
for production use

Qwen-VL

✓ SELECTED

Reliable Arabic OCR
with vision-language support

Qwen-VL was the only model that handled **Arabic legal text consistently** without hallucinating critical fields like numbers, dates, and names.

CORE — MODEL SELECTION

Picking a Qwen-VL Version

KEPT

Qwen 2.5VL

Stable vision-language model with solid Arabic OCR baseline. Available in 3B parameter size.

KEPT

Qwen 3VL

Newer generation with improved document understanding and stronger multilingual vision capabilities. Available in 4B and 8B sizes.

CORE — MODEL SELECTION

Size Variants per Version

Qwen 2.5VL

3B PARAMS

Qwen 3VL

4B PARAMS	**8B** PARAMS

3 total candidates trained and evaluated — the best selected by **human evaluation** on test results.

CORE — TRAINING

From Candidates to Final Model

Unsloth
Training framework

→

LoRA Fine-Tuning
Applied to all 3 size variants

→

Human Evaluation
Judged by observing test output quality

→

Qwen 3VL — 4B
Best overall OCR accuracy on Arabic contracts

Winner: Qwen 3VL 4B — selected by human judges for highest accuracy on Arabic legal documents.

RAG SYSTEM

Why We Need RAG

PURPOSE 01

Article Retrieval

CAT has a small context window — RAG finds exactly which article the user's prompt refers to and passes only that to CAT.

PURPOSE 02

Metadata Consistency

When CORE converts a contract, RAG links each article to its related metadata field — so CAT edits stay consistent with the full JSON structure.

RAG — PURPOSE 01

RAG for Article Retrieval

THE PROBLEM

A contract can be 20+ pages with dozens of independent articles. CAT's context window is too small to receive the whole file — and it doesn't need to.

20+ pages

50+ articles

Context overflow

CAT fails

WITH RAG

RAG reads the user's prompt, finds the exact article being referenced, and returns only that article's ID — so CAT receives just what it needs.

RAG Pipeline

- User prompt received — Step 1
- Matching article found — Step 2
- Article ID → CAT — Step 3

RAG for Metadata Consistency

THE PROBLEM

Some articles are directly linked to metadata fields like `company_capital`. If CAT edits that article, the metadata becomes stale and inconsistent.

Inconsistency Risk

- × Article 7 edited → mentions new capital
- × `company_capital` field still shows old value

WITH RAG

On first CORE conversion, RAG scans every article, detects which metadata field it relates to, and adds a reference attribute — so any future CAT edit automatically updates the linked field.

Enriched Article JSON

```
"id": "7",
"value": "رأس المال...",
"metadata_ref": "company_capital"
```

RAG — ARCHITECTURE

Models & Mathematical Foundation

DENSE RETRIEVER

Qwen3-embedding-0.6B

Converts Arabic contract clauses into dense semantic vectors. Stored and searched using **FAISS** for fast nearest-neighbour lookup.

Embedding FAISS Index

↓

RE-RANKER

Qwen3-reranker-0.6B

Cross-encoder that re-ranks the top retrieved clauses by evaluating the query and clause **jointly**, boosting precision beyond embedding search alone.

Cross-Encoder Top-K Rerank

① Embed Clauses

$$v_i = f(d_i)$$

Each clause d_i is passed through the embedding model f to produce a dense vector v_i

② Normalize

$$\hat{v}_i = v_i / \|v_i\| \quad \hat{q} = q / \|q\|$$

Both clause vectors and query vector are L2-normalized before comparison

③ Similarity & ④ Top-K Retrieval

$$S(q, d_i) = \hat{q}^T \hat{v}_i \rightarrow R = \mathrm{TopK}(\, S(q, d_i)\,)$$

Cosine similarity between query and each clause; top-K highest scores retrieved from FAISS

⑤ Re-Rank & ⑥ Best Clause

$$r_i = g(q, d_i) \rightarrow d^* = \mathrm{argmax}(r_i)$$

Cross-encoder g scores each (query, clause) pair jointly; the highest-scoring clause d* is selected

THE AI ENGINE

Why CAT & CORE Instead of General AI?

FEATURE / CATEGORY	GENERAL AI (CHATGPT / GEMINI)	CAT & CORE
Privacy & Security	✗ External cloud processing	✓ Local office deployment
Domain Knowledge	✗ General purpose only	✓ Accounting & contract expertise
Document Understanding	✗ Entire contract context	✓ Clause-level retrieval (CORE)
Editing Precision	✗ May rewrite unrelated content	✓ Targeted, precise amendments
Operational Cost	✗ Pay per request / subscription	✓ One-time deployment cost
Consistency	✗ Output may vary	✓ Standardized amendment process

GRADUATION PROJECT · 2026

Thank You

Built with care, trained with precision, and deployed with purpose — CLARA is ready.

THE TEAM

AK	**Ahmed Elkot**	Project idea & CAT model	CAT
AY	**Ahmed Yehia**	CORE model	CORE
SM	**Saif Mohamed**	RAG & Logs feature	RAG
AY	**Amr Yasser**	CAT score function & HR helper	SCORE · HR
MA	**Mohamed Anwar**	Task management & file organization	Tasks · Files
AE	**Adham Eihab**	Security & contract output formatting	Security

ME SUPERVISOR
Dr. Mohamed El Halaby

Questions welcome